MARK SCHEME · TEACHER COPY

11BR · Topic 3: Hardware

Cambridge AS Level Computer Science 9618

Paper 1 Theory Fundamentals

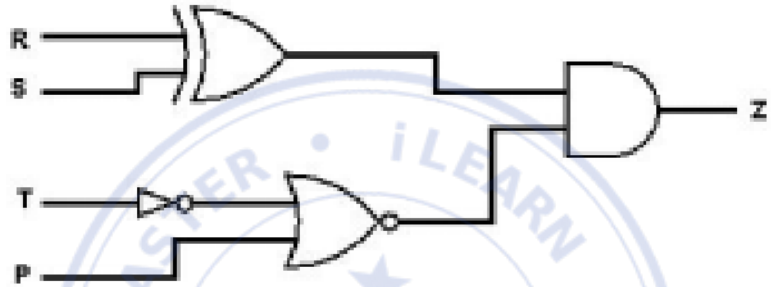
Total marks: 75

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

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Question 1 · May/June 2023 Paper 12, original Q6 (a),(b) [4 marks]

Question	Answer	Marks
1(a)	**1 mark** for correct XOR and AND gates, **with** correct inputs **1 mark** for correct NOT and NOR gates **with** correct inputs 	2
1(b)	**1 mark** for first 4 rows **1 mark** for last 4 rows	2

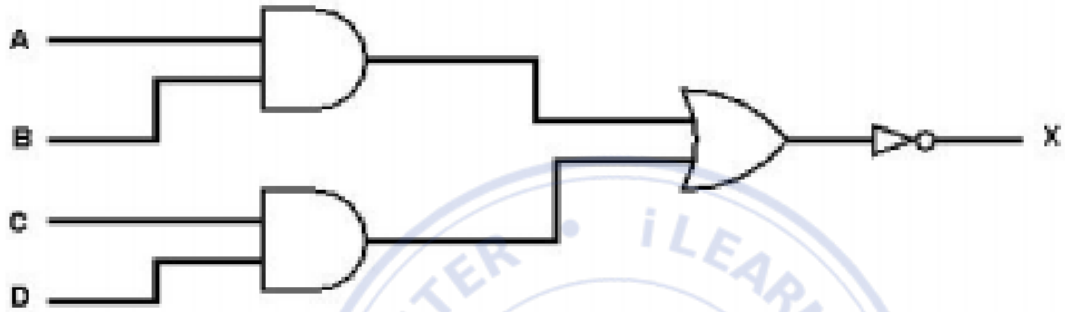
P	Q	R	Working space	Z
0	0	0		0
0	0	1		1
0	1	0		1
0	1	1		1
1	0	0		1

Total: 75 marks

1	0	1		0
1	1	0		1
1	1	1		1

Question 2 · May/June 2024 Paper 12, original Q1 (a),(b) [6 marks]

Question	Answer	Marks
2(a)	**1 mark** for each correct answer: *NAND* • The output is 0 when both inputs are 1, otherwise the output is 1 *NOR* • The output is 1 when both inputs are 0, otherwise the output is 0 *XOR* • The output is 1 when one of the inputs is 1 and the other input is 0, otherwise the output is 0 *OR* • The output is 0 when both inputs are 0, otherwise the output is 1	**4**

2(b)	**1 mark** for both AND gates with correct inputs **1 mark** for correct OR and NOT gates with correct inputs and no superfluous gates:  	**2**

Question 3 · May/June 2025 Paper 12, original Q4 (a),(b) [4 marks]

Question	Answer	Marks
3(a)	**1 mark** each: • X = (R NOR S) XOR T • Y = (T AND U) OR U	**2**
3(b)	**1 mark** for first four rows, **1 mark** for second four rows	**2**

A	B	C	Working space	X
0	0	0		1
0	0	1		1

0	1	0		**1**
0	1	1		**1**
1	0	0		**1**
1	0	1		**0**
1	1	0		**0**
1	1	1		**1**

Question 4 · May/June 2025 Paper 13, original Q4 (a),(b) [4 marks]

Question	Answer	Marks
4(a)	**1 mark** for each correct statement X = B NOR (A NOR C) Y = (B AND C) XOR NOT D	**2**
4(b)	**1 mark** for 1st 4 rows **1 mark** for 2nd 4 rows	**2**

A	B	C	Working space	X
0	0	0		**1**
0	0	1		**1**

0	1	0		**1**
0	1	1		**0**
1	0	0		**0**
1	0	1		**1**
1	1	0		**1**
1	1	1		**1**

Question 5 · May/June 2026 Paper 13, original Q8 (a),(b),(c) [10 marks]

Question	Answer	Marks	Guidance
5(a)	**1 mark** for name, **1 mark** for a matching description • EPROM • Erase using UV rays // needs removing to erase // have to erase whole device // supports many read/write cycles • EEPROM • Erase using electric signals // erase in situ // can update parts of memory // uses floating gate technology // supports many read/write cycles	**4**	No mark for expanding acronym for description.
5(b)	**1 mark** each. • The microphone has a diaphragm / ribbon • The incoming sound **waves** cause **vibrations** of the diaphragm • … causing a coil to move past a magnet // causing a magnet to move past a coil (dynamic microphone). • // changing the capacitance (condenser microphone) // deforms the crystal (crystal microphone) • An electrical signal is produced	**3**	For MP1 accept diaphragm equivalent e.g. cone.

5(c)	**1 mark** each • Temporary storage of data • …until the device is ready to receive it // until a sound player takes the data • Data may be sent faster than it can be received // Allows for different speed of transmission between devices • Allows processor to carry out other tasks	**3**	

Question 6 · Oct/Nov 2023 Paper 12, original Q1 (a),(b),(c) [6 marks]

Question	Answer	Marks
6(a)	**1 mark** for: • infra-red / proximity (sensor)	**1**
6(b)	**1 mark** for each bullet point (**max 2**) • Actuator generates a signal / causes an action / converts electrical energy into a mechanical force • ... to push an arm // to open a trap door // to pick up the chocolate bar with the incorrect weight	**2**
6(c)(i)	**1 mark** for each feature (**max 2**) • Dedicated to a single task // limited number of functions • Built into a larger system // integrated into a larger system • An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	**2**

6(c)(ii)	**1 mark** for each bullet point (**max 1**) • Difficult to change / update the firmware by the user // Difficult to upgrade devices to take advantage of new technology • Cannot be easily **adapted** for **another** task • Troubleshooting faults/upgrading is a specialist task • Devices are often thrown away rather than repaired as difficult to upgrade or repair	**1**

Question 7 · Oct/Nov 2023 Paper 12, original Q4 (a),(b) [5 marks]

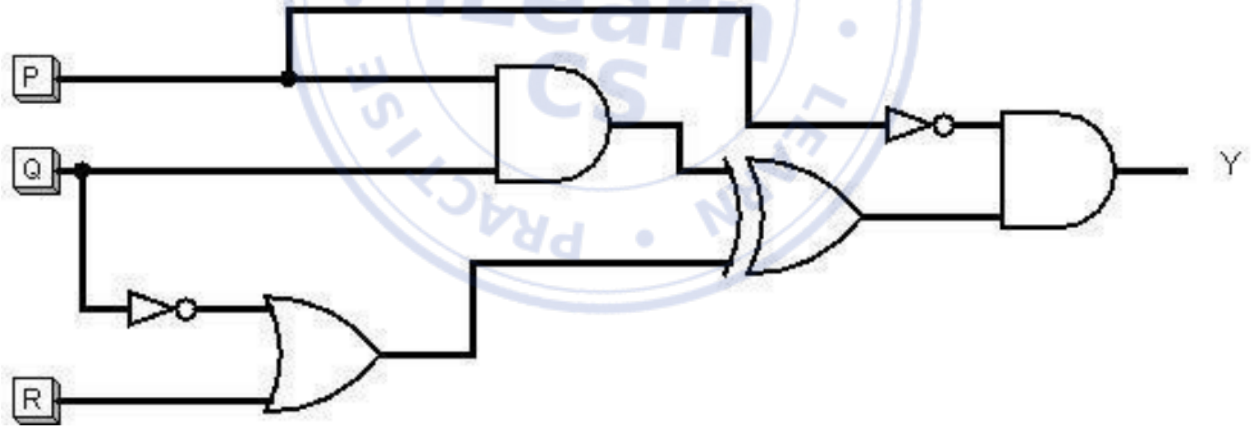
Question	Answer	Marks
7(a)	**1 mark** for each bullet point • A NAND B • NOT(B XOR C) • Final NAND X = A NAND B NAND (NOT(B XOR C))	**3**

7(b)	**1 mark** for each set of rows (shaded)	**2**

A	B	C	X
0	0	0	**0**
0	0	1	**0**
0	1	0	**1**
0	1	1	**0**
1	0	0	**1**
1	0	1	**1**
1	1	0	**1**
1	1	1	**1**

Question 8 · Oct/Nov 2023 Paper 13, original Q4 (a),(b) [4 marks]

Question	Answer	Marks

8(a)	**1 mark** for each set of highlighted rows **P** | **Q** | **R** | **Y** 0 | 0 | 0 | 1 0 | 0 | 1 | 1 0 | 1 | 0 | 0 0 | 1 | 1 | 1 1 | 0 | 0 | 0 1 | 0 | 1 | 0 1 | 1 | 0 | 0 1 | 1 | 1 | 0	**2**
8(b)	**1 mark** for **P** AND **Q** and NOT **Q** OR **R** **1 mark** for NOT gate and AND gate and XOR gate 	**2**

Question 9 · Oct/Nov 2023 Paper 13, original Q7 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks

9(a)	**1 mark** for each bullet point (**max 2**). • Static RAM has faster access time • …because it does not need to be refreshed • …used on the CPU for improvement of CPU cache speed • Static RAM has lower	**2**
9(b)	**1 mark** for each bullet point (**max 2**) • EEPROM allows frequent / multiple read / write / erase operations • …which means that the contents of the firmware in the headset can be changed easily • …without fully erasing the contents of the firmware in the headset first	**2**
9(c)	**1 mark** for each bullet point (**max 4**) • The print instructions and data are sent by the laptop to a buffer (at laptop speed) • The data is transferred from the buffer to the printer (at printer speed) • …allowing user to continue using the laptop // and allowing processor to continue processing • …instead of waiting for relatively slower printer • When the buffer is empty an interrupt is sent to the laptop • …requesting more data	**4**

9(d)	**1 mark** for naming a correct port **2 marks** for matching justification • USB • …has fast data transfer speeds for data (to the headset) • …is a universal/popular cable // universal standard • HDMI • …allows video and audio to be transferred (on the same cable) • …convenience of HDMI as no need for two cables	**3**

Question 10 · Oct/Nov 2024 Paper 12, original Q2 (a),(b),(c) [6 marks]

Question	Answer	Marks
10(a)	**1 mark** for each bullet point (**max 3**) e.g. • It is difficult to change / update the firmware by the user • Errors cannot be fixed easily // Troubleshooting / fault-finding / repairing is a specialist task / expensive • Functionality cannot be changed/extended easily // Cannot be easily adapted for another task • Faulty / outdated devices are often thrown away rather than repaired • … leading to e-waste	**3**

10(b)	**1 mark** from: • DRAM is less expensive to manufacture/purchase than SRAM in the embedded system • DRAM has a higher bit density per chip // more data can be stored per chip	**1**
10(c)	**1 mark** for each bullet point (**max 2**) • EPROM uses ultraviolet light to erase data whilst EEPROM uses an electrical signal to do this • EPROM has to be removed from the circuit board when changing the data whilst EEPROM remains in the circuit when the data is changed • EPROM erases all the data, EEPROM can erase parts of the data	**2**

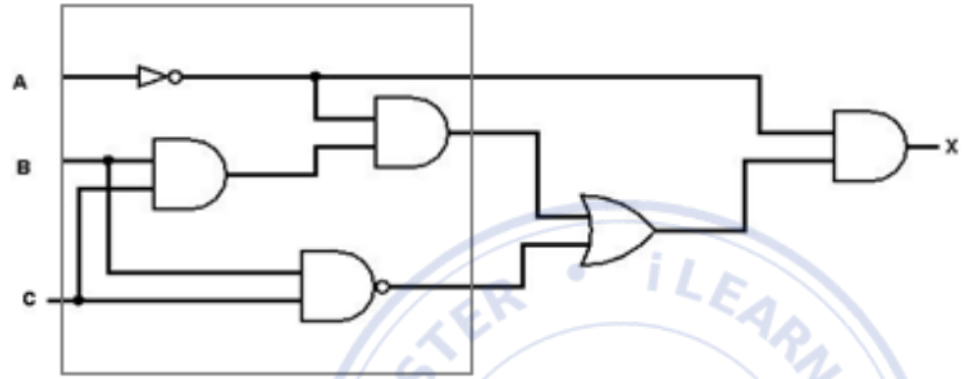
Question 11 · Oct/Nov 2024 Paper 12, original Q9 (a),(b) [4 marks]

Question	Answer	Marks
11(a)	**1 mark** for sensor **and** its use in this system • Infra-red sensor • Measure / check the height of the vehicle • Pressure sensor • Measure / check the weight of the vehicle	**2**

11(b)	**1 mark** for each bullet point (**max 2**) for a correct justification No marks for the identification of the system Monitoring system • Because there is no use of feedback // the warning sign is only an indicator • ... the output of the turning on of the sign does not affect the input of data from the sensors • The system does not have any actuators	**2**

Question 12 · Oct/Nov 2024 Paper 13, original Q1 (a),(b),(c) [5 marks]

Question	Answer	Marks
12(a)	**1 mark** for: Output is only 1 if one input is 1 and the other is 0 // Output is only 1 if both inputs are different // Output is only 0 if both inputs are the same	**1**
12(b)	**1 mark** each: • ((**A** NOR **B**) AND (**A** OR **B**)) • NAND (**B** AND **C**) **X** = ((**A** NOR **B**) AND (**A** OR **B**)) NAND (**B** AND **C**)	**2**

12(c)	**1 mark** for NOT gate, 2 correct AND gates and the NAND gate **1 mark** for OR gate and the final AND gate 	**2**

Question 13 · Oct/Nov 2024 Paper 13, original Q2 (a),(b) [6 marks]

Question	Answer	Marks
13(a)	**1 mark** for each difference e.g. • DRAM requires constant refreshing, but SRAM does not need to be constantly refreshed • DRAM generally has greater capacities per chip, but SRAM generally has lower capacities per chip • DRAM has slower access times than SRAM	**2**

13(b)	**1 mark** for each bullet point (**max 4**) • The computer and the hard disk drive transmit and receive at different speeds // The computer transfers data faster than the HDD can receive • The buffer is used for temporary storage • … so that the computer can transfer data to the buffer at the higher speed • ... and is not held up waiting for data to transfer • … and so that data is transferred to hard disk drive from the buffer at the slower rate	**4**

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